Huawei Access Node Optical Transport Failure Resolution SOP (WFD-ACC-09)

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8631

Description

The Optical Transport Failure on Access Node WFD-ACC-09 is attributed to the degradation of SFP/XFP modules, fiber cuts, or power loss in the receive (Rx) or transmit (Tx) paths. This failure domain is specific to the Access role on Huawei equipment due to its reliance on Optical Transport Network (OTN) technology for data transmission. OTN utilizes a series of wavelength division multiplexing (WDM) lasers to transmit data at various wavelengths, which are then received and processed by the SFP/XFP modules. The Huawei Access node, in particular, employs a Huawei proprietary OTN protocol stack to manage OTN frame processing and transmission. The failure of SFP/XFP modules or fiber cuts can disrupt this process, leading to the ALARM_8631 Major alarm. In addition, power loss in the Rx or Tx paths can also contribute to this failure domain. It is essential to diagnose and resolve this issue promptly to ensure uninterrupted data transmission and prevent potential network congestion.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that the necessary equipment is within reach.

- Check the redundant link status to ensure that the network remains operational in the event of a failure.

- Gather necessary safety equipment and follow proper safety procedures when working with electrical and optical equipment.

Potential Root Causes

Cause	Probability
SFP/XFP module degradation	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display otu info, Expected Output: OTU information, Description: This command displays the OTU information, including the wavelength, transmit power, and receive power. It helps identify any power loss or degradation in the Rx or Tx paths.

Command: display otu error, Expected Output: OTU error information, Description: This command displays the OTU error information, including the error count, error rate, and error cause. It helps identify any errors in the OTU frame processing and transmission.
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```
Command: display interface brief, Expected Output: Interface brief information, Description: This command displays the interface brief information, including the interface status, transmit rate, and receive rate. It helps identify any issues with the interface or the SFP/XFP module.
```

Resolution Steps

Step 1: Step: Swap the SFP/XFP module, Command: swap otu-sfp xfp , Description: Swap the SFP/XFP module with a new one. This may resolve the issue if the degradation is caused by a faulty module.

Step 2: Step: Check and clean the fiber, Command: display fiber info, Description: Check and clean the fiber to ensure it is free from any debris or contamination. This may resolve the issue if the fiber cut is caused by a physical obstruction.

Step 3: Step: Adjust the Rx/Tx power, Command: set otu tx-power && set otu rx-power , Description: Adjust the Rx/Tx power to ensure it is within the acceptable range. This may resolve the issue if the power loss is caused by a configuration error.

Post-Check Verification

```
Command: display otu info, Expected Output: OTU information, Description: Verify that the OTU information is within the acceptable range and that there are no errors in the OTU frame processing and transmission.

Command: display interface brief, Expected Output: Interface brief information, Description: Verify that the interface brief information is within the acceptable range and that there are no issues with the interface or the SFP/XFP module.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: swap otu-sfp xfp , Description: Restore the original SFP/XFP module to its previous state. This may be necessary if the swap operation did not resolve the issue.

- Step: Restore the original fiber configuration, Command: undo set otu tx-power && undo set otu rx-power , Description: Restore the original fiber configuration to its previous state. This may be necessary if the power adjustment operation did not resolve the issue.